



Graphs in Machine Learning

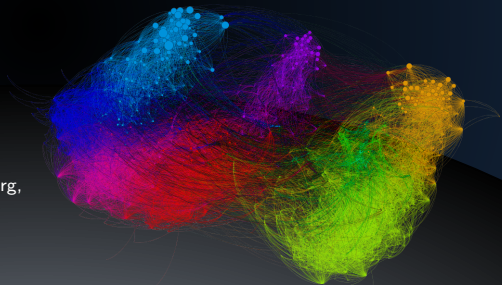
Spectral Clustering: Graph Cuts

MinCut, RatioCut, and NCut

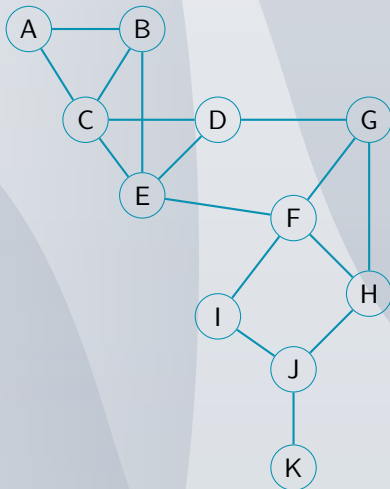
Michal Valko

Isara Labs

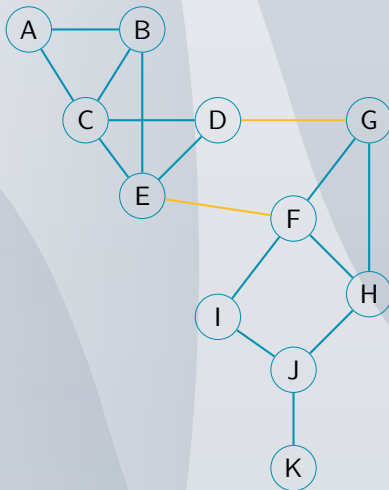
Partially based on material by: Ulrike von Luxburg,
Gary Miller, Doyle & Schnell, Daniel Spielman



Spectral Clustering: Cuts on graphs

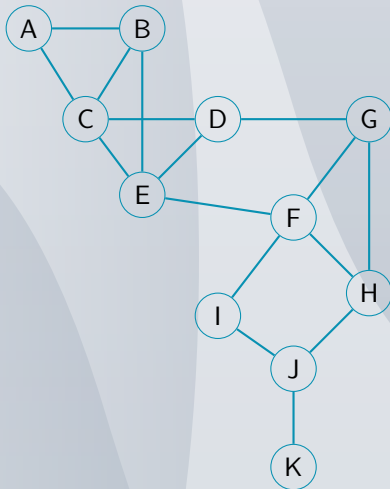


Spectral Clustering: Cuts on graphs



Defining the cut objective we get the clustering!

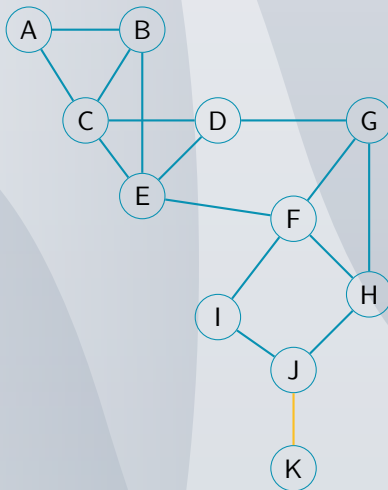
Spectral Clustering: Cuts on graphs



MinCut: $\text{cut}(A, B) = \sum_{i \in A, j \in B} w_{ij}$

Are we done?

Spectral Clustering: Cuts on graphs



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Are we done?

Can be solved efficiently, but maybe not what we want

Spectral Clustering: Balanced Cuts

Let's balance the cuts!

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Normalized Cut

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Easily generalizable to $k \geq 2$

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Can we compute this? RatioCut and NCut are NP hard :(

Approximate!

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<https://misovalko.github.io/mva-ml-graphs.html>